TEC: Enhanced Error Correction for Noisy Teleported States under the Depolarising Noise Model

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1 Abstract

Quantum teleportation is susceptible to noise and decoherence, causing transferred states to no longer be true representatives of the original quantum information. We study a teleportation based error correction (TEC) technique using three qubit repetition codes and stabilisers to protect the quantum information during the teleportation process. The density matrix simulations are performed in Qiskit, where we compare the baseline teleportation protocol to TEC both with and without corrections on the stabiliser measurements. Under depolarising noise, TEC reduces the logical error rates up to 64% compared with the baseline, whereas uncorrected stabilisers are shown to degrade the fidelity of the fully optimised TEC implementation. This is also demonstrated with the time evolution of the Lindblad equation to extend the analysis to hardware noise, where prolonged circuit duration exposes qubits to greater environmental decoherence. The results determine TEC as a practical tool for improving the fidelity, and therefore reduce logical errors in quantum communications subject to noise.

2 Introduction

2.1 Quantum teleportation

Quantum communication was first proposed in 1993 by Bennett et al. [1]. The paper introduced the idea of transferring an unknown state from one party, conventionally referred to as Alice, to another some distance away, called Bob. Importantly, the state itself does not undergo spatial translation during the process. Central to the protocol is quantum entanglement, a non classical phenomenon in which the two subsystems share a joint quantum state. The result of measuring one subsystem instantly determines the correlated outcome of the other, independent of their spatial separation [2]. Refer to appendix 7.1.1 for an introduction to the concepts of entanglement. Alice must perform a local measurement, resulting in the collapse of the superposed state into a definite outcome accessible to Bob. The no-cloning theorem, a fundamental consequence of quantum mechanics, dictates that it is impossible to create an identical copy of a quantum state and also physically transfer it elsewhere [3]. Therefore, Bob alone cannot obtain any meaningful result from his state's random outcome without some information from Alice. Quantum teleportation consequently requires working in both quantum and classical domains. For Bob to be able to reconstruct the original state, a classical communication channel is intrinsically required to convey the outcome

of the Bell measurement [4]. This classical transmission enforces causality and prevents faster than light signalling, ensuring Einstein's theory of special relativity is not violated [5].

The standard teleportation protocol operates using the transmission of two classical bits for a three qubit system. This system consists of the unknown state held by Alice to send to Bob, and Alice's and Bob's respective qubits in the shared entangled Bell pair [6]. Alice performs a Bell basis measurement between her entangled pair and the state unknown to her, which destroys the state locally. The collapse of the states' wavefunction results in one of the four Bell pair outcomes, $|\psi^+\rangle$, $|\psi^-\rangle$, $|\phi^+\rangle$, $|\phi^-\rangle$, which Alice communicates to Bob by sending two classical bits via the classical channel. Once Bob obtains Alice's transmitted message, he can apply one of the four Pauli corrections (X, Y, Z, I), representing specific single qubit unitary transformations, to his qubit [2, pp. 25–28]. In doing so, he can reconstruct the original quantum state. Note that the classical bits do not contain any quantum information, but rather it is the protocol's setup of shared entanglement between the two parties that allows for the state to be reconstructed by Bob. The accuracy of the reconstructed quantum state is assessed using the fidelity, F, defined as a measure of the agreement between the original and reconstructed state [4]. Unit fidelity ($F = 1$) corresponds to a perfect reconstruction of the original state. Refer to the appendix 7.1 for the baseline circuit setup for teleportation and its mathematical details.

Quantum mechanics is a complementary tool when integrated within technological frameworks that already rely on well established classical communication. Their combination offers distinct advantages that cannot be obtained through classical methods alone [7]. A key interest that drives research is the use of cryptography (QKD) and the idea of unbreakable data security in forming future quantum networks [8]. Traditional encryption methods, which are dependent on computational hardness and public-key systems, are expected to be undermined in the future by the computational power of quantum computers [10]. These quantum systems have the capability of efficiently solving classical problems, most notably through algorithms such as Shor's [11]. Quantum teleportation is a primitive for building quantum networks that will enable the secure transfer of data [9]. Experimental demonstration of teleportation is currently being researched across a range of physical platforms using superconducting qubits, photons, trapped ions, and solid-state systems [12]. Research aims to improve fidelity and extend transmission distances to implement within emerging quantum communication networks.

The specific objective of this paper is to use TEC to improve the fidelity of

the quantum teleportation platform.

2.2 Limitations to quantum teleportation systems

Open and closed systems: When describing an ideal quantum system, it must evolve by some unitary operator, U: $|\psi\rangle \rightarrow U|\psi\rangle$, such that the state transforms deterministically and without being corrupted [13],[14]. Within this framework, the teleportation protocol can be modelled as a sequence of gate operations acting on an entangled multi-qubit system[2, pp. 25–27]. The absence of errors in the system results in a perfect reconstruction of the state. In practice, however, the teleportation protocol cannot be treated as a closed system when considering realistic implementations [15]. The qubits within the protocol are coupled to an environment, which forces the system to evolve as an open system, and can be represented by the density matrix ρ [16]. Formally, we can describe this by assuming the system and environment states are initially separable [2, pp. 93–94]. The relation for separability is $\rho \otimes \rho_{env}$ [2, pp. 357], where the unitary transformation between their density matrices becomes an entangled joint system $U(\rho \otimes \rho_{env})U^{\dagger}$ [17]. After interacting with the environment, the final state is given by, $\epsilon(\rho) = Tr_{env}[U(\rho \otimes \rho_{env})U^{\dagger}]$ [2, pp. 359], such that the map is no longer unitary (pure state). The resulting state becomes mixed and cannot be reversed, reflecting the loss of information to the environment, which we refer to as noise [18]. Noise reduces the fidelity of the protocol by unwanted interactions with the environment's degrees of freedom that corrupt the quantum state before Bob receives it from Alice [19].

One significant challenge with noise in the teleportation circuit is the waiting times between gate operations [20]. The duration of the circuit operations allows the environment to interact with the idle qubits, such that as time evolves, the qubit experiences decoherence. Decoherence is the process by which the qubit becomes irreversibly entangled with the environmental degrees of freedom, thereby driving the density matrix, ρ to relax towards a statistical mixture [21]. Refer to appendix 7.1.2 for the formalisms on how decoherence affects the density matrix. Understanding decoherence is crucial for scalable quantum systems where only a subset of qubits can be operated on at once [22]. The question then lies in how long a qubit can reliably hold a quantum state before the environment maximally mixes it and becomes severely degraded or unusable. This motivates the study of noise models and error correction for teleported states under open systems.

With these concepts established, the paper is structured in the following manner. Section 3 introduces the two complementary noise models, depolarising

and Lindblad, that will be applied to the input state before the teleportation protocol. Section 4 will overview the QEC techniques for stabilisers and three qubit bit-flip codes used for TEC. The section 4.3 will describe how TEC is complemented with QEC and the limitations of correcting certain types of errors. Section 5 provides the analysis of whether TEC improves the fidelity in comparison to just a noisy teleportation circuit. Section 6 finalises the discussion of TEC fidelity and possible future works.

3 Noise Implementations on Teleported States

3.1 The Lindblad master equation

The Lindblad master equation, based on Born-Markov approximations, describes the evolution of an open system's density matrix, ρ, when it interacts with a weakly coupled environment [23],[24]. The equation shows the decoherence of the qubit by the errors it accumulates between gate operations at a rate of γ over some time t:

$$\frac{\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \Sigma_k \gamma_k \Big(L_k \rho L_k^\dagger - \frac{1}{2}\big\{ L_k L_k^\dagger, \rho \big\} \Big) \tag{1}$$

here $\frac{1}{2}\big\{ L_k L_k^\dagger, \rho \big\}$ is the dampening term, where the curly brackets denote the anti-commutator $\{X, Y\} = XY + YX$. The term ensures trace-preserving (positive) evolution. Additionally, $L_k \rho L_k^\dagger$ is the jump term, which contains the set of operators L_k that describes how the environment acts on the system by what noise is induced. If there is no noise acting, the von Neumann equation $\frac{\rho}{dt} = -\frac{i}{\hbar}[H, \rho]$ is recovered, where the Hamiltonian, H, depicts the (ideal) system's unitary evolution [25].

3.2 The depolarising channel (CCPT)

In this report, the depolarising model, $\epsilon(\rho) = \frac{I}{2}p + (1-p)\rho$ [26], is selected as the type of preexisting noise exclusively induced onto the three qubit input state before it is teleported in the protocol. The parameter p represents the probability that a state undergoes a uniformly distributed random Pauli error in the Bloch sphere. A quantum state is said to be maximally mixed under noise when its density matrix is $\frac{I}{2}$ [2, pp. 378]. In the depolarising channel, this maximum decoherence ($p = 1$) occurs when the output state is $\epsilon(\rho) = \frac{I}{2}$. Otherwise, the state is pure if there is no noise, $\epsilon(\rho) = (1-p)\rho$. The depolarising noise expressed in Kraus form is written using the Pauli matrices and in terms of the parameter p as follows:

$$\epsilon(\rho) = \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) + (1-p)\rho \tag{2}$$

where the Pauli matrices X, Y, Z are Hermitian, $X^\dagger = X$ and are applied with equal probability of $\frac{p}{3}$. The channel map can be represented with the Kraus operators $\epsilon(\rho) = \sum_k E_k \rho E_k^\dagger$ [27]. The set of Kraus operators, $\{E_k\}$, satisfies $\sum_k E_k E_k^\dagger = 1$ for the trace to be preserved and positive.

This has geometrical consequences on the Bloch sphere as the Bloch vector, $\vec{r}$ of any input state, under the Pauli error form, uniformly decreases towards the centre by a multiple of $(1 - \frac{4}{3}p)$. Every state is therefore treated the same by virtue of its SU(2) symmetry [2]. Crucially, it has been selected so that the parameter, γ or decoherence rate, is the same regardless of the Pauli error type on the qubit, $L_1 = X, L_2 = Y, L_3 = Z$ and $\gamma_k = \gamma_1 = \gamma_2 = \gamma_3$.

3.3 Hardware realism from the theoretical depolarising noise model

To benchmark the performance of TEC against realistic noise, the depolarising channel will be selected as the primary noise model. The corresponding fidelity, F between the ideal and noisy states is state independent [28], and given by the equation:

$$F = 1 - \frac{2}{3}p \tag{3}$$

where p is the depolarising probability. In order to connect the theoretical noise model to a physically motivated noise process, we will parametrise p using the analytical solution of the Lindblad master equation for Markovian decoherence. For a qubit decay of rate, γ, the depolarising probability evolves as follows:

$$p(t) = 1 - e^{-4\gamma t} \tag{4}$$

This allows us to sweep over circuit time, t, at fixed noise rates, γ, and observe how the parameter p of the depolarising channel affects the degradation of the qubits in the protocol.

The aim is to emulate hardware realistic noise in the circuit whilst maintaining errorless gate operations and evaluating whether the TEC protocol protects the transmitted qubits. This will be done by applying identity gates

on the three encoded input qubits, before transmission, and mapping the chosen noise channel, ε to the identity gate type within Qiskit's Aer noise model.

The Aer framework allows gate level noise to be assigned to each independently affected qubit, so that the total noise is given by $\varepsilon(\rho) = (\varepsilon \otimes \varepsilon \otimes \varepsilon)\rho_{in}$. Using this controlled method is necessary to determine any improvement in the fidelity from implementing TEC. The gate operations are chosen to be ideal, and noise does not act on the entangled Bell pair or during the teleportation. Furthermore, we used the density matrix backend for the noise to be mapped as positive trace preserving, following the Kraus form discussed in section 3.2.

3.3.1 Noise mitigation on Markovian vs non-Markovian processes

There are many methodologies in ongoing research for how to mitigate or correct noise [29]. In this report, we will focus on the quantum error correction (QEC) techniques modelled for Markovian processes. Particularly on their integration into the teleportation error correction (TEC) framework [30].

It is crucial to note that real quantum devices deal mainly with non-Markovian behaviour where noise is not memoryless [31]. On the contrary, the original theoretical formulation of QEC was developed under Markovian noise models. Such assumptions, which rely on time local master equations, do not capture the memory effect observed in contemporary real quantum devices. Nevertheless, the methodology of TEC to correct noise could potentially be extended to non-Markovian processes and provide future aid in quantum networks.

4 QEC Framework for TEC Implementations

The following section will describe the full implementation of TEC onto the built teleportation circuit. We will operate using eleven physical qubits distributed across four registers, including ancilla qubits used for syndrome extraction. The work is based on the methodology presented in the "Demonstration of teleportation based error correction in the IBM quantum computer" paper [30]. The separation of these blocks allows for each stage to be analysed independently, so that errors are contained and fixed before propagating further into the system.

4.1 Three qubit bit-flip codes

Logical Bell pair encoding (I): The entangled Bell pair between Alice and Bob is prepared at the start of the teleportation protocol before any operation occurs during circuit runtime. This is deliberately done to use the pair as an error filtering channel when the unknown state held by Alice, that have noise applied, interacts with it. Preparing it first ensures it has not been exposed to errors, so it is a clean, entangled resource where the complex amplitudes and general structure of the pair have no uncertainty.

Each logical qubits $|000\rangle$ and $|111\rangle$ in the Bell pair is encoded with three physical qubits, resulting in a six qubit entangled state shown below [29].

$$\frac{|00\rangle_I + |11\rangle_I}{\sqrt{2}} \rightarrow \frac{|000000\rangle + |111111\rangle}{\sqrt{2}} \tag{5}$$

logical qubits encoding (II): The input state unknown to Alice is sampled uniformly from the Bloch sphere using the following equation:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \tag{6}$$

where the polar angle, $\theta \in [0, \pi]$ and the azimuthal angle, $\phi \in [0, 2\pi]$ [32]. A uniform Haar measure is important for the teleportation to behave trivially for both $|0\rangle_I$ and $|1\rangle_I$, and ensures the fidelity degradation of the state can be calculated. We store the state as a density matrix for the evolution of noise. The single qubit input states are subscripted with I to denote them as logical qubits which encode the data that will be transmitted to Bob. The information held by the three physical qubits is encoded into the three qubit bit-flip code using two CNOT gates as follows:

$$|\psi\rangle_I = \alpha|0\rangle_I + \beta|1\rangle_I \rightarrow \alpha|000\rangle + \beta|111\rangle \tag{7}$$

All three physical qubits $|q_0, q_1, q_2\rangle$ have the same computational value, and the complex amplitudes α and β are preserved in the superposition state by not being measured throughout the encoding.

The conceptual framework for repetition codes works as follows. Let q_0 be the data qubit with the actual information $|q_0\rangle_I$ in the three qubit unknown state $|\psi\rangle_I$. Furthermore, let the 'empty' ancillary qubits be q_1 and q_2, which are prepared as $|0\rangle$. Later on, the ancilla qubits are used for the redundancy. The combined state is given by the equation below:

$$(\alpha |0\rangle + \beta |1\rangle) \otimes |0\rangle \otimes |0\rangle = \alpha |000\rangle + \beta |100\rangle \tag{8}$$

where two CNOT gates are applied, CNOT 1: $\alpha |000\rangle + \beta |110\rangle$ and CNOT 2: $\alpha |000\rangle + \beta |111\rangle$. The encoded logical state is then $\alpha |0\rangle_I + \beta |1\rangle_I$, where $|0\rangle_I = |000\rangle$ and $|1\rangle_I = |111\rangle$.

This process of encoding is called the redundancy and follows a majority vote system. If a single qubit flips due to a Pauli-X error, the other two qubits remain with the original value, such that the error can be identified by which qubit disagrees out of the three. This detection method allows us to recover the original logical information despite some physical qubits being corrupted. The majority vote error is complemented using stabilisers Z_0Z_1 and Z_1Z_2 for reliable error correction, refer to section 4.1.1 below.

Basis measurement for phase-flip errors (III): The three qubit code's fundamental restriction lies in there being no account for phase-flip Pauli-Z errors. The state may show no bit-flip errors and pass the state as errorless, e.g., the complex amplitudes phase $e^{i\pi} = -1$ changes (phase-flip): $\alpha |000\rangle - \beta |111\rangle$. To extend protection against phase-flip errors, Hadamard gates, H will be applied to all three encoded input data qubits before and after the noise region by exploiting basis change measurements [2, Sec. 10.5.3]. Using the basis change trick $HZH = X$, it converts any phase-flip that occurs in the depolarising noise channel into a bit-flip in the rotated basis, making it detectable and correctable by the same Z stabilisers that work with bit-flips.

Entangling data with Bell pair (IV): Required for the teleportation protocol to exist as a three qubit joint system, a CNOT gate is applied between each qubit in the unknown state and its corresponding qubit in Alice's Bell pair. This is important as the information encoded in the input state is no longer localised on the data qubits alone, but distributed across the Bell pair. The distribution permits the subsequent state resulting from Alice's Bell measurements to be Pauli corrected by Bob using the classical bits transmitted through the channel.

4.1.1 Stabilisers measurements

The conceptual aim of TEC is to extract the error on the qubits before Bob makes his final correction on his state. If a Bell measurement is performed between Alice and the unknown state, the joint system that contains the error collapses into one of the four classical Bell state outcomes. Measuring this way permanently destroys the superposition causing the information

containing the error to become unrecoverable. In contrast, stabilisers make non-destructive measurements without collapsing Alice's qubits by using parity checks. The stabilisers used are the product of two Pauli operators Z that act on adjacent qubits. They are given by $S_1 = Z_0 Z_1$ and $S_2 = Z_1 Z_2$. The eigenvalues of the stabiliser are $+1$ if both adjacent qubits hold the same value $Z|0\rangle = |0\rangle$, or -1 if they are not the same, $Z|1\rangle = -|1\rangle$ [29]. However, stabilisers cannot decipher phase-flip errors, as they only perform parity checks on the qubits. This the reason for the use of Hadamard gates introduced in part (III).

Ancillary qubits for the stabilisers (V): In the circuit, each stabiliser S_1 and S_2 is measured using an ancilla qubit as a non-destructive measurement tool. The two ancilla qubits in the 11-qubit system, q_{10} and q_{11}, are prepared by applying a Hadamard gate, H on the initial qubit state $|0\rangle$, such that there is a superposition of states represented by $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$. Both ancilla qubits are reset before every round of measurements to avoid preexisting errors being stored. In doing so, the system is modelled as a Markovian process.

The process for stabiliser $S_1 = Z_0 Z_1$ and its corresponding ancilla qubit q_{10}, in the full three qubit state (q_0, q_1, q_{10}), proceeds as follows. Two CNOT gates are applied to each of the control qubits (q_0, q_1) to the corresponding target ancilla qubit q_{10}, forming the three qubit state $\frac{|000\rangle+|111\rangle}{\sqrt{2}}$. A second Hadamard gate is then applied on the ancilla q_{10}, forming the state below:

$$\frac{1}{2}|0\rangle(|00\rangle + |11\rangle) + \frac{1}{2}|1\rangle(|00\rangle - |11\rangle) \tag{9}$$

Only the ancilla qubit is measured, so it is q_{10} that collapses whilst the data qubits q_1 and q_2 are untouched. The outcome of measuring the ancilla reveals the parity associated with the stabiliser S_1. This preserves the error information contained within the data qubits, rather than being lost upon measurement. If the ancilla qubit collapses to $|0\rangle$, no error has occurred, where $q_0 = q_1$, corresponding to the parity $Z_0 Z_1 = 1$. In contrast, if the ancilla collapses to $|1\rangle$, an error has occurred since $q_0 \neq q_1$, resulting in an odd parity stabiliser value $Z_0 Z_1 = -1$. The process above follows the same logic for stabiliser $S_2 = Z_1 Z_2$.

4.2 Syndrome extraction and stabiliser based correction

Collapsing ancillas for syndrome diagnosis (VI): The three repetition code measures the stabilisers to detect single qubit errors. There exists four possible syndrome outcomes that diagnose the corresponding error, summarised by the table 1 below:

Error on qubit	(S_1, S_2)	(s_1, s_2)
No error	$(+1, +1)$	00
Error on q_0	$(-1, +1)$	10
Error on q_1	$(-1, -1)$	11
Error on q_2	$(+1, -1)$	01

Table 1: The possible syndrome outcomes determine the qubit that is corrected, either with an X gate for bit-flip errors or a Z gate for phase-flip errors.

Stabiliser based correction (VII):

Once the syndrome is obtained, we can correct the error based on the diagnosis, presented in table 2. For bit-flip noise errors, X or Z gates are applied to the qubit indicated by the syndrome. This portrays the majority vote rule of the three qubit repetition code and ensures that any single qubit error is corrected before teleportation.

Syndrome (s_1, s_2)	Pauli Correction
00	I
01	X or Z on q_2
10	X or Z on q_0
11	X or Z on q_1

Table 2: Syndrome based Pauli correction applied to the encoded block.

4.3 The TEC Protocol

4.3.1 Corrections by the teleportation protocol

After the stabiliser corrections are applied on Alice's side, the logical state is sent to Bob via the teleportation method. For each of Bob's encoded qubits i, two classical syndrome bits are produced based on the Bell measurement performed by Alice.

Across the three encoded qubits, this yields six classical bits that are stored in the register. Each bit contains the Pauli frame that Bob must apply to recover the teleported state, where the mapping is shown in Table 3. For each physical qubit i, the even indexed bit $[2i]$ determines whether a Pauli-X correction is required, while the odd indexed bit $[2i+1]$ determined whether a Pauli-Z correction is required.

$m = (\text{X-bit}, \text{Z-bit})$	Bob's correction
$(0, 0)$	I
$(1, 0)$	X
$(0, 1)$	Z
$(1, 1)$	XZ

Table 3: Pauli corrections are applied by Bob based on the Bell measurement using classical syndrome bits defining the Pauli frame.

After Bob's correction, the teleportation based error correction protocol is completed. To keep Bob's correction fault-tolerant in the teleportation, we performed traversal operations by acting independently on corresponding physical qubits without entangling across the encoded block. Most importantly, the teleportation is not deterministic at the quantum level, but only when the classical information is used.

4.3.2 Fidelity calculations

For TEC to be demonstrated as a useful error correction tool, we must use the fidelity to measure how close the reconstructed state is to the original after teleportation.

Ideal logical encoding (VIII): Bob's three qubit state is projected onto the logical subspace to be decoded and renormalised, so it is a valid density matrix:

$$\{|000\rangle, |111\rangle\} \longrightarrow \{|0\rangle, |1\rangle\} \tag{10}$$

By constructing an ideal decoder that has no syndrome extraction, we assumed that there is noiseless decoding on Bob's three qubit state. As a result, Bob's correction after the teleported state can be measured with fidelity in the Pauli frame.

Instead of saving one averaged density matrix, ρ, and computing the fidelity, we simulated one density matrix per classical outcome.

This yields a set of density matrices $\rho_B^{(m)}$ for Bob's three qubit encoded state. Each $\rho_B^{(m)}$ corresponds to the quantum state of Bob given that the classical outcome was m. Without doing classical conditioning, all outcomes would be incoherently mixed and cause the phase information to be destroyed even when the noise is zero ($p = 0$). Computationally, this is what the classical channel does in the teleportation protocol. The ideal decoder explained in (VIII) is then applied to $\rho_B^{(m)}$ to obtain the decoded density matrix, $\rho_{decoded}^{(m)}$ for each outcome m.

The computed fidelity must be in the Pauli frame $X^x Z^z$. This is done to ensure that when $p = 0$, the fidelity starts at $F \approx 1$, and the phase information is conserved. For each classical outcome, m, the output state, corresponding to the classical bit sent by Alice, is $X^x Z^z |\psi\rangle$ [35]. The Pauli frame is not noise, but deterministic by the value m. The fidelity is then computationally calculated with the following equation:

$$F_m = F(X^x Z^z |\psi\rangle \rho_{decoded}^{(m)}) \tag{11}$$

where F_m is the fidelity of outcome m. Fidelity is then averaged for all the outcomes.

The combination of three qubit bit-flip codes, stabiliser diagnosis, and teleportation, allows for any single qubit error to be detected and corrected in anticipation of Bob receiving the state.

4.3.3 TEC limitations

The implemented TEC in this report applies error correction in two passes rather than a single unified quantum error correction code. Referring back to 4.1, the first pass corrects bit-flip errors (X) using the Z-basis stabiliser measurements. The second pass detects and corrects phase-flip errors (Z) by rotating the encoding basis with Hadamard gates before and after the noise region. This allows the stabiliser circuit to correct the noise. The two passes are applied sequentially on the same set of qubits. As the two passes are treated independently, the Pauli error Y is not reliably corrected.

It is important to highlight that the TEC approach is not a complete quantum error correcting code (QECC). It is not equivalent to a nine qubit code like the Shor code, which is designed to correct all single qubit Pauli errors simultaneously, nor is it completely fault-tolerant.

Despite these limitations, the sequential approach suits the objectives for a

few reasons. The main goal is to demonstrate that TEC can improve the fidelity of a teleported state during transmission and not the construction of a complete fault-tolerant error correcting code. The intended purpose of the qubits is to survive decoherence during the teleportation process, which is finite, instead of being protected against an accumulation of errors over time.

4.3.4 Lindblad parameters for IBM like hardware implementations

Each physical qubit is modelled as a superconducting qubit consistent with IBM implementations. To align our depolarising model with the hardware, we chose to reinterpret the Lindblad decay rate as the dephasing rate $\frac{1}{T_\phi}$, and the evolution time as the teleportation circuit time, t. The decoherence of superconducting qubits are described by the timescale equation below:

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi} \tag{12}$$

where T_2 is the total coherence time, T_1 is the energy relaxation, and T_ϕ is the dephasing time [2, sec. 8.3.5]. We assumed that the error dominating our decoherence is T_ϕ, due to flux noise coupling to the environment when qubits are idle in between operations in the teleportation. The longer the circuit cycle, the greater the noise the qubits experience. For the Markovian pure dephasing time expressed by the Lindblad operator $L = \sqrt{\gamma Z}$, the decoherence is shown by $T_\phi = \frac{1}{2\gamma}$ [16]. The evolution is equivalent to the Pauli-Z channel with probability as follows:

$$p_Z(t) = \frac{1}{2}\left(1 - \exp\left(-\frac{t_{\text{TEC}}}{T_\phi}\right)\right) \tag{13}$$

Due to the noise depolarising channel, with rate γ being equivalent for all Pauli errors, our chosen gamma for the superconducting qubits is $\gamma = \frac{t}{T_\phi}$.

5 Analysis of Selected Error-correction Techniques on Teleported States

5.1 The analysis approach

The analysis in the following section will be conducted from a computational angle. The two selected models for these correction techniques are TEC with depolarising noise and hardware representative Lindblad. We will initially introduce the baseline for the simple case of three qubit encoded teleportation as the framework for building a TEC protocol that improves fidelity. All the computational analysis on our built TEC structure will be compared to the provided baseline.

5.2 The baseline: noisy teleportation states with no TEC

We first establish the performance of standard teleportation under depolarising noise as a comparative reference for all TEC results. The (Pauli) fidelity, F is plotted against the depolarising model probability values, p between the range (0-0.30), as shown below in figure 1.

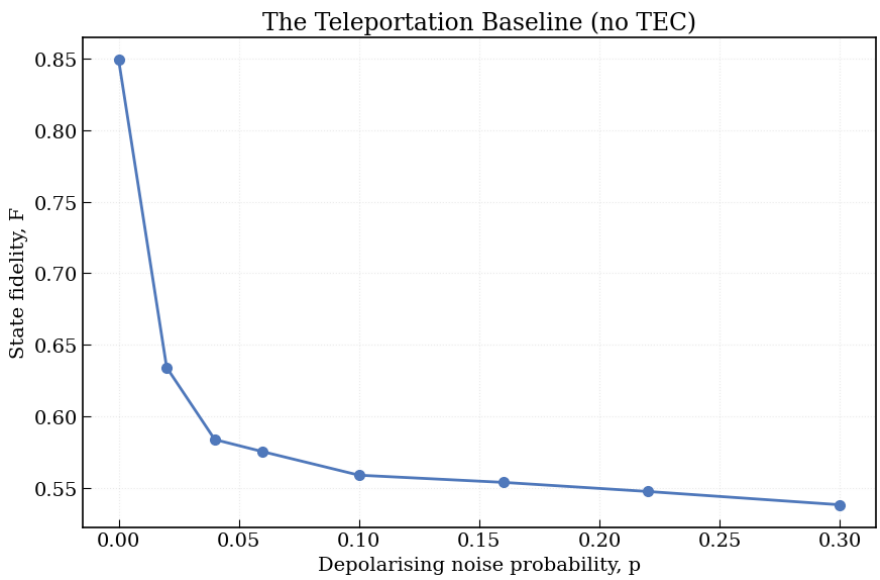

Figure 1: Baseline three qubit teleportation: The fidelity, F against the depolarising probability, p , where fidelity is in the Pauli frame.

From the result above, the fidelity curve demonstrates a linear decay due to the physical depolarising noise. The most prominent fidelity drop lies within the region of $F \approx 0.85$ to $F \approx 0.64$, up to the point where $p \approx 0.02$. From $p \approx (0.02 - 0.3)$, the fidelity linearly decreases at a slower rate, stopping at $F \approx 0.54$. At small values of p, the depolarising noise already introduces phase errors that the repetition code cannot correct, as described in section 4.1.

An important distinction for TEC implementation later on is that our teleportation used is not unitarily preserving fidelity at $p = 0$. This is due to our decoding procedure used in (VIII), and not a failure of the protocol itself. When the decoder projects the logical qubit to a physical one, the information encoding the error is lost. Therefore, when the density matrix, $\rho_{decoded}$, is renormalised, the fidelity is reduced.

Any TEC improvements on the fidelity are evaluated relative to this baseline, where fidelity starts at $F \approx 0.85$. The reason for a loss of fidelity at $p = 0$ is explained later.

5.3 PART I: The TEC analysis

5.3.1 The approach

In the section 4.3.3, we highlighted the limitations of our constructed TEC when combating different Pauli noise types from the subjected depolarising channel. Whether TEC improves the fidelity of the noisy teleported states is addressed in a structured manner. The analysis in section 5.3.2 begins by examining bit-flip and phase-flip errors in the teleportation protocol independently.

We proceeded to examine, in section 5.3.3, the scenarios of TEC with and without correction on the stabilisers, compared against the no TEC case on noisy teleported states. The aim is to evaluate if, at any point, our circuit depth exposes our qubits to additional noise and, as a consequence, degrades the fidelity. Refer back to the limitations of quantum systems 2.2 for the description of open systems.

The graphs in this section plot eight depolarising probability, p points in the range $p = (0 - 0.3)$.

5.3.2 TEC tackling errors: bit-flip and phase-flip Pauli errors

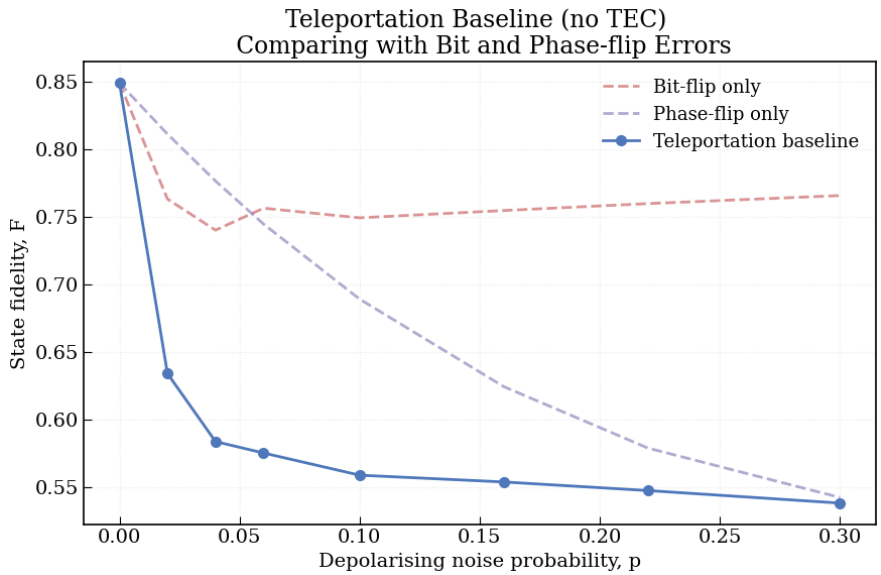

Figure 2: A direct comparison of the baseline with bit and phase-flip errors on the graph Fidelity, F against a varying depolarising noise probability, p.

The baseline curve at small p has an initial drop between $F \approx 0.85$ and $F \approx 0.64$, and then slowly decreases linearly before flattening near $F \approx 0.55$ at $p = 0.30$. Our teleportation curve exhibits a statistical mixture of Pauli errors due to the depolarising channel model in section 3.2.

From the bit-flip curve, the fidelity initially drops to $F \approx 0.74$, and then gradually recovers itself at larger values from p ≈ 0.04 onwards. This is due to the Pauli frame structure explained in section 4.3.1.

The phase-flip curve shows the fidelity immediately decaying monotonically, similar to the behaviour of our teleportation baseline plot. The distinction between the bit-flip curve flattening and the phase-flip curve decaying is that the decoder cannot recover the loss of relative phase imposed by the Pauli-Z error. Such effects on the baseline, as a result of Z-type and Y-type errors, show to dominate at small values of p before X-type errors can begin to improve the fidelity.

Implementing TEC is therefore a key requirement to detect and address the Pauli-Z errors that in part cause the baseline to decay.

5.3.3 TEC scenarios

We progress to implementing the full TEC onto our teleportation circuit by adding the stabilisers mentioned in the previous section 4. The two scenarios of TEC with and without correction on the stabilisers, were compared against no TEC. The goal is to distinguish whether there are fidelity improvements from using stabilisers to address the Pauli-Z errors described in section 5.3.2.

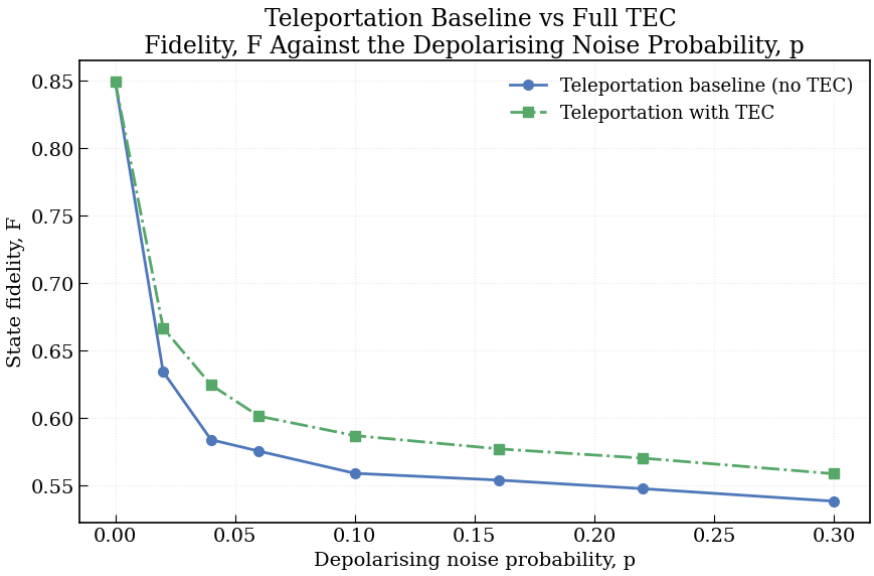

Figure 3: Fidelity, F against the depolarising probability, p for the two scenarios 1) No TEC (standard teleportation protocol), and 2) full TEC.

Before implementing TEC, we demonstrated from our baseline cure, that repetition codes could not reduce the monotonic decay of the fidelity. The figure 3, indicates that in the immediate affect, $p \leq 0.02$, of noise being introduced, the TEC curve decays slower and maintains fidelity higher than no TEC. Within a low noise region, each 1% increment of noise reduces the teleportation fidelity by approximately $(4-5)\%$ relative to our ideal fidelity, $F \approx 0.85$. At higher noise, $p > 0.02$, both curves decay monotonically. Notably, TEC remains consistently above the baseline at roughly a $(5-8)\%$ relative fidelity improvement. The TEC performance decreases at higher depolarising probabilities $p \approx 0.30$ and onwards as the curves flatten.

In the simulation, starting at the depolarising probability $p = 0$ means that there is no depolarising noise or Pauli-type errors. Implying, the density matrix must be exact and the protocol has ideal gate operations. The reason why figure 3 displays the fidelity, $F < 1$ at $p = 0$ is not based on physical

noise, but due to the intrinsic structure of the teleportation protocol. Before decoding, Bob's three qubit state due to noise is not necessarily the original teleported state. The logical teleportation contains many measurement conditioned Pauli frames, which our fidelity is averaged over all the classical branches after decoding. Projection of the repetition code results in the loss of fidelity as explained in section 5.2.

This demonstrates that TEC can overall improve the fidelity up until the Pauli-Z and Y errors dominate at larger polarising probabilities, p.

5.3.4 Stabiliser corrections and the trade-off in fidelity

Additionally, fidelity for TEC in absence of stabiliser corrections was intentionally analysed to examine whether background operations reduce the benefit of implementing TEC entirely. This comparison accesses if the fidelity improvements from TEC are from measuring stabilisers or correcting them.

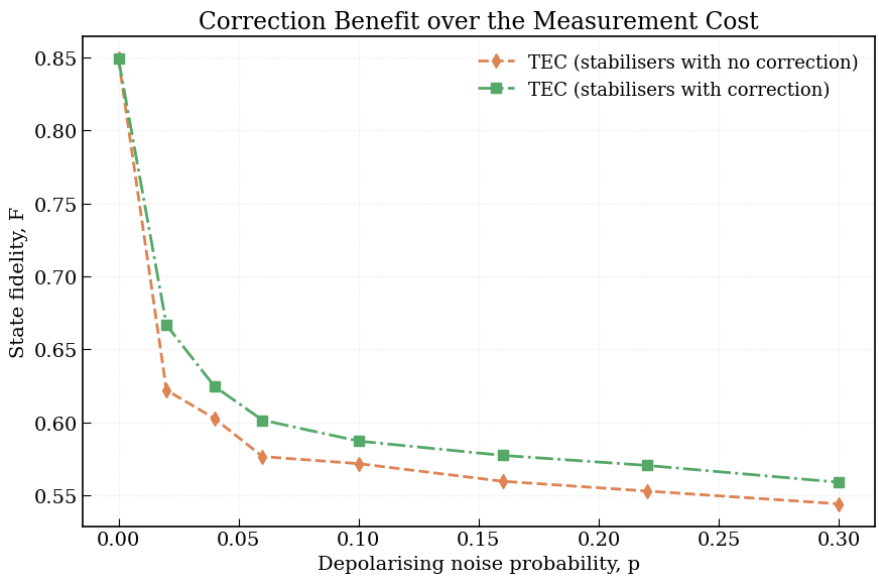

Figure 4: Fidelity, F against the depolarising probability, p for the two scenarios 1) TEC with no correction on the stabilisers, and 2) TEC with correction on the stabilisers.

Figure 4 reveals the trade-off implemented in the error correction of stabilisers. Stabilisers introduce decoherence when not corrected, despite their measurements providing information about physical errors. As a result, this

effect lowers the fidelity, shown by the no correction curve monotonically decreasing at a faster rate. The full TEC protocol is consistently above the no correction curve, highlighting that actively correcting detected errors sustains a higher fidelity than the baseline. Therefore, the benefit of stabilisers is only apparent if the information is used, otherwise their existence in the protocol outweigh any benefit to the fidelity of our teleported state.

5.3.5 Logical error rates vs the depolarising noise

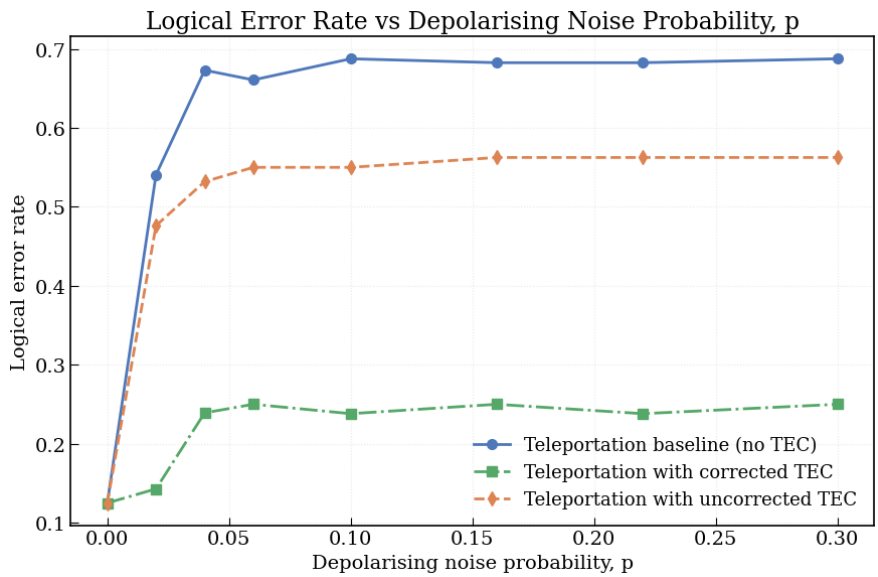

Figure 5: The logical error rate against the depolarising noise probability p for the three teleportation scenarios: 1) No TEC (standard teleportation protocol), 2) TEC with no correction on the stabilisers, and 3) TEC with correction on the stabilisers.

In the absence of correcting errors, our baseline shows logical errors that dominate towards 70% at very low values of p when single qubit errors begin to accumulate. The result of not decoding the logical state is shown from the uncorrected TEC curve, revealing a 21% improvement from our baseline. In comparison, the teleportation with correction significantly suppresses the errors by bit-flip corrections by 64% relative to the baseline. The three curves flatten approximately at $p = 0.16$ reflecting the presence of uncorrectable errors which three qubit bit-flip codes cannot combat. This is important as figure 5 demonstrates the TEC protocol actively suppresses errors and not just improves the fidelity.

5.3.6 Further improvements to combat error types

As discussed in section 4.3.3 a complete quantum error correcting code like the nine qubit Shor code would address the limitations our TEC protocol has in correcting Pauli-Y errors. Our current TEC only works if we correct bit and phase-flip errors independently. However, encoding each logical (data) qubit across nine physical qubits would require a significant number of more qubits than the eleven used in our built TEC. This would also increase the circuit depth massively. Shor code is a natural extension of the presented work, and its integration into the teleportation framework is left as a direction for future studies.

5.4 PART II: The Lindblad description of TEC

The structure of TEC remains the same, however we replace the depolarising noise model with Lindblad dynamics. Refer back to section 3.3 for the specific parameters. Under the Lindblad model, circuit time, t describes how long the idle qubits experience decoherence as explained in section 2.2. Graphs in this section plot for 25 depolarising probability, p points.

5.4.1 Using TEC to reduce the decoherence of states during teleportation circuit time

The figure 6 shows the fidelity, F against the circuit time for the baseline and full TEC case.

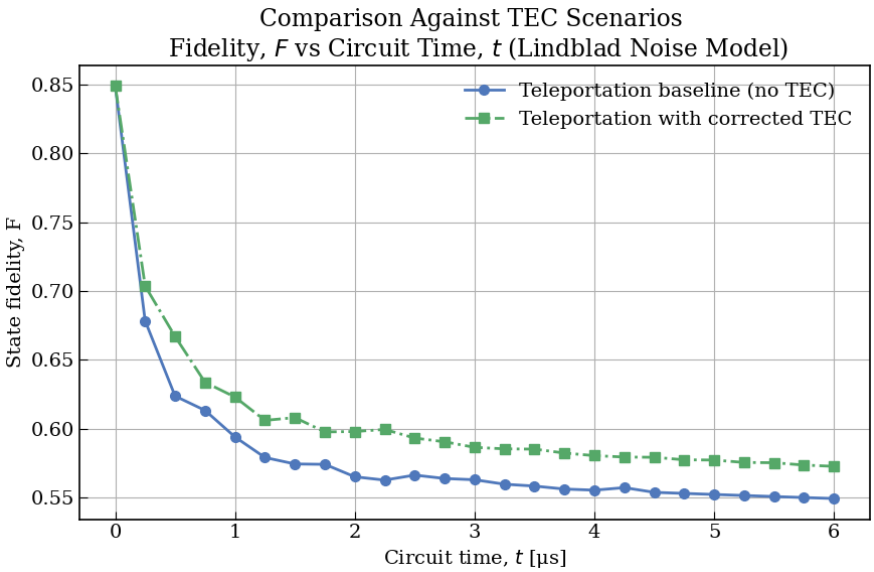

Figure 6: Fidelity, F against the circuit time, t for the two scenarios 1) The TEC baseline, and 2) TEC with correction on the stabilisers.

As the circuit time, t increases, the Lindblad noise acting on the encoded state decreases the fidelity. TEC reveals to maintain approximately $(4-5)\%$ higher fidelity throughout the circuit time, whilst both curves present a monotonic decay. Furthermore, the most dominant reduction is $< 1\,\mu$s the time range, where the gain in fidelity is 5%. The coherence time of the qubits is extended in the teleportation protocol by using TEC.

5.4.2 Logical error rates on the Lindblad noise model

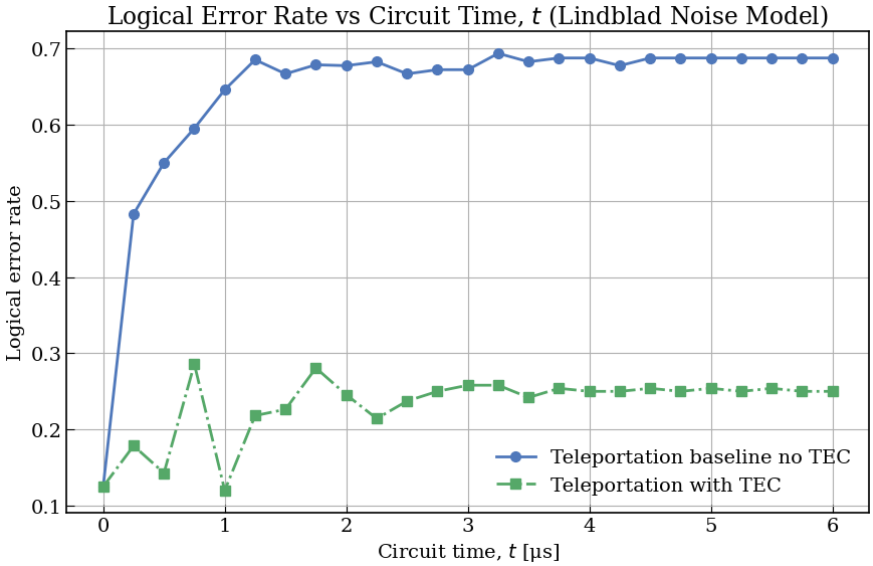

Figure 7: The logical error rate against the circuit time, t for the two teleportation scenarios: 1) No TEC (standard teleportation protocol), 2) TEC with correction on the stabilisers.

To extend logical error rates to realistic noisy hardware by using the Lindblad model, plot 7 was produced. Similarly to figure 5, TEC shows to suppress the logical errors on our qubits as the circuit time, t increases. Within the circuit time region $t = (0 - 2.2)\,\mu$s, produce a non-monotonic curve due to the interaction between the Lindblad decoherence and stabiliser corrections. In this low decoherence region, the overall trends shows that bit-flip errors can be corrected immediately after the syndrome is extracted. At longer circuit time approximately $t \approx 2.2\,\mu$s, the uncorrectable Pauli errors start to dominate. The curve monotonically increases slightly and then flattens around $t \approx (3.7 - 6)\,\mu$s.

6 Conclusions

We investigated the ability of TEC to protect the decoherence quantum information in the presence of depolarising and Lindblad noise. Three qubit bit-flip codes with stabilisers measurements were implemented using density matrix simulation in Qiskit, allowing a systematic comparison between the baseline encoded teleportation, syndrome extraction without correction, and

TEC with stabiliser corrections. Moreover, the effect of independent bit-flip and phase-flip errors were also analysed to isolate where the degradation of qubits were occurring.

Under the depolarising noise, the teleportation protocol showed to rapidly accumulate logical errors as the depolarising probability, p increased. This lead to a reduction in the Pauli reconstructed fidelity and a high logical error rate on the qubits. An important result of this report is the requirement for error correcting our stabilisers to prevent the degradation of TEC's ability to improve the fidelity of the teleported states. For the full TEC, logical error rates were suppressed by 64% relative to the baseline teleportation. The findings demonstrated that the primary benefit of TEC is from actively correcting errors rather than from stabilisers measurements alone.

We extended the analysis to a hardware representative Lindblad noise model and increased the teleportation circuit time to determine whether the decoherence of the qubits caused by the environmental interactions, are reduced by the new TEC implementations. The Lindblad TEC fidelity showed to slow down the decoherence extending the lifetime of the teleported state within the circuit duration. This was shown by the TEC fidelity maintained approximately 5% above the teleportation baseline and the improvement was stable over the circuit time up to $6\,\mu$s.

The study reveals that teleportation error correction (TEC) can improve quantum teleportation against noise. Our findings naturally inline with the current research done in experimental quantum platforms like superconducting qubits or ion traps. TEC could become a useful tool for scalable quantum communication and computing systems.

7 Appendix

The section constructs the theoretical framework required to understand quantum teleportation under depolarising noise.

7.1 The basics of the teleportation protocol

The initial setup between two parties, Alice and Bob, begins with three qubits, the intended unknown state to be teleported, $|\psi\rangle$ held by Alice, and the pair of entangled qubits q_1 and q_2 for Alice and Bob, respectively. Additionally, two classical bits (0 and 1's) are used to communicate between Alice and Bob through a classical channel.

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|0\rangle \tag{14}$$

$$|\alpha| + |\beta| = 1 \tag{15}$$

For creating an entangled pair between Alice and Bob, the process requires the application of quantum gates, specifically Hadamard, H, and CNOT gates. The H gate transforms a simple basis state, either $|0\rangle$ or $|1\rangle$ of Alice's qubit into the superposition of both states with amplitude $1/\sqrt{2}$ as follows.

$$H\,|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = H \otimes I\,|00\rangle \tag{16}$$

For both qubits to be entangled, measuring either Alice or Bob must give sufficient information to know the other qubits state. In the above case, measuring Bob's qubit, q_2, to be always $|0\rangle$ does not give any information on Alice's qubit being $|0\rangle$ or $|1\rangle$. Applying a CNOT gate will allow the pair to be entangled, also referred to as an entangled Bell state. The overall process of creating the bell pair $|\psi^+\rangle$:

$$|00\rangle \xrightarrow{H\otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{CNOT} \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{17}$$

Note that there exists four Bell basis in total that can express the combined state of Alice and Bob's qubit after the quantum gate operations, $|\psi^+\rangle$,$|\psi^-\rangle$,$|\phi^+\rangle$,$|\phi^-\rangle$. The important concept is that the entangled pair is a joint system such that it shares the same wavefunction. When either qubit is measured, they both collapse to either $|0\rangle$ or $|1\rangle$ as the system is in superposition.

The total system in the protocol is expressed such that after some bra-ket rules and applying the Pauli matrices, the superposition of the unknown state and the entangled qubit pair before measurement is as follows:

$$|\psi\rangle \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{18}$$

$$= \frac{1}{2}\Big\{ |00\rangle\,(\alpha\,|0\rangle + \beta\,|1\rangle) + |01\rangle\,(\alpha\,|1\rangle + \beta\,|0\rangle) + |10\rangle\,(\alpha\,|0\rangle - \beta\,|1\rangle) + |11\rangle\,(\alpha\,|0\rangle - \beta\,|1\rangle) \Big\} \tag{19}$$

$$= \frac{1}{2}\left\{ |00\rangle |\psi\rangle + |01\rangle X |\psi\rangle + |10\rangle Z |\psi\rangle + |11\rangle XZ |\psi\rangle \right\}. \tag{20}$$

When Alice makes a measurement, the global superposition collapses to one of the four outcomes. Suppose that it collapses to $|01\rangle$, then Alice sends two classical bits "10" such that Bob knows Alice collapsed to $(\alpha |0\rangle - \beta |1\rangle)$ and he must then apply the Pauli Z matrix to recover the unknown state $|\psi\rangle$.

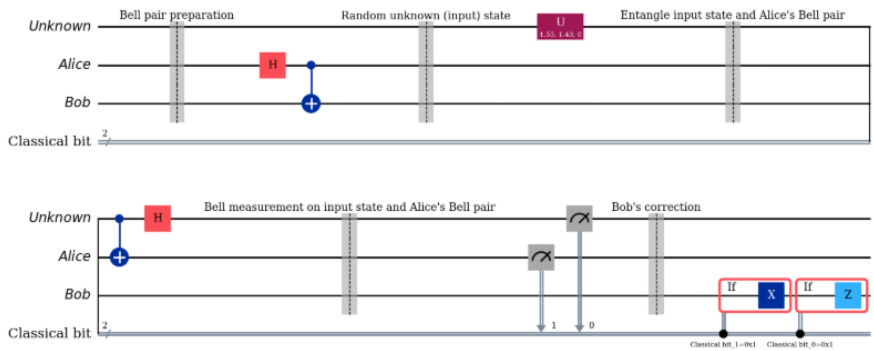

Figure 8: The teleportation circuit: Alice sends the unknown state to Bob through their shared quantum entanglement by a Bell measurement and passing the classical bits through the classical channel.

7.1.1 Entanglement

The joint system of two qubits, q_1 and q_2 has the following wave-function:

$$|\psi\rangle = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \sigma |11\rangle \tag{21}$$

A non-entanglement state is one that can be written as the tensor product of two wavefunctions $|\psi\rangle \otimes |\phi\rangle$, where the two qubits have their own wave-function [2, sec. 2.4.1]. Measuring one does not affect the other and this is referred to as separable states.

$$\begin{aligned} &(\alpha |0\rangle + \beta |1\rangle) \otimes (\gamma |0\rangle + \sigma |1\rangle) \\ &= \alpha\gamma |00\rangle + \alpha\sigma |01\rangle + \beta\gamma |10\rangle + \beta\sigma |11\rangle \end{aligned} \tag{22}$$

The condition for separable sates is as follows, otherwise the two qubits are entangled.

$$\begin{aligned} \alpha\gamma &= \alpha \\ \alpha\sigma &= \beta \\ \beta\gamma &= \gamma \\ \beta\sigma &= \sigma \end{aligned} \tag{23}$$

This implies the condition for separability $ad = bc$, otherwise the state is entangled as it cannot be written as a product.

7.1.2 Density matrix formalism

The state vector $|\psi\rangle$ provides a direct description of a known quantum state or formally named as a pure state. The motivation for density matrix arises due to the two different types of uncertainties: classical uncertainty whether a state is one or the other, and the quantum superposition of both or mixed state. The density matrix is a mathematical operation that describes the framework of mixed and pure states in a unified way. Considering a general quantum system with i states each with probabilities, p_i, the density matrix, ρ is described by the summation $\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|$, [34]. A single pure state described in the teleportation protocol section, the density matrix takes the form:

$$\rho = |\psi\rangle \langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \tag{24}$$

The diagonal entries $|\alpha|$ and $|\beta|$ determine the probability of measuring state 0 or 1, and the off-diagonal determine the relative phase difference/coherence of the two states. Decoherence decays the off-diagonal terms $\rho_{01} = \alpha\beta^*$ and $\rho_{10} = \alpha^*\beta$ linearly with the noise strength, p, while preserving the classical measurement probabilities $\rho_{00} = |\alpha|^2$ and $\rho_{11} = |\beta|^2$. As p increases, the state transitions from a pure state when $tr(\rho^2) = 1$ to a mixed state when $tr(\rho^2) < 1$. This shows how decoherence destroy quantum information by eradicating the phase relationship.

7.2 Simulations Software Environment

The simulations were run on Qiskit version 2.3.0 and Python version 3.13.9.

7.3 Simulations Code Repository

The simulation code is stored in the GitHub public repository link below.

`https://github.com/sandraNSan/Teleportation-Error-Correction.git`

7.4 Computing hardware and runtimes

All the simulations were performed on a Windows 11 PC equipped with an Intel i5-7500 (4 cores, 3.4GHz) processor and 16GB of RAM memory. The simulation utilised approximately 13.5GB of RAM memory. The TEC simulation execution times required 85 minutes to complete for eight points, while the Lindblad simulations were completed in 121 minutes for 25 points.

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